

Lecture - 2

Linear Equations

In this Lecture, we will solve a system of two equations having two variables

$$ax + by = f$$

$$cx + dy = g$$

a, b, c, d, f and g are numbers.

* Before we calculate the solutions, we observe that for any numbers $\alpha \neq 0, \beta \neq 0$, $ax + by = f$ is equivalent to $\alpha ax + \alpha by = \alpha f$. Similarly, $cx + dy = g$ is equivalent to $\beta cx + \beta dy = \beta g$.
(Symbol for equivalence $\sim$)

For example

$$2x + 3y = 6 \sim 10x + 15y = 30$$
$$x - y = 3 \sim 4x - 4y = 12$$

Solution We shall break up the solution into several cases.

Note that the system may have only one solution, no solutions or infinitely many solutions (this happens if the second equation is equivalent to the first)

Case 1 $a=d=0, c \neq 0, b \neq 0$

Then, the System of equations becomes

$$y = \frac{f}{b} \text{ and } x = \frac{g}{c}$$

Thus, the only solution is $(\frac{g}{c}, \frac{f}{b})$.

Note: A Similar argument works for $c=d=0, a \neq 0, b \neq 0$.

Case 2 $a \neq 0, c \neq 0$ and $ad-bc \neq 0$

First we multiply $aX + bY = f$ by $1/a$ and obtain the equivalent equation

$$X + \frac{b}{a}Y = \frac{f}{a}.$$

Now we multiply the above equation by c and obtain the equivalent equation

$$cX + \frac{bc}{a}Y = \frac{fc}{a}.$$

Thus our system of equations become

$$cX + \frac{bc}{a}Y = \frac{fc}{a}$$

$$cX + dY = g$$

Now we subtract these two equations and obtain

$$\left(\frac{bc}{a} - d\right)Y = \frac{fc}{a} - g.$$

Hence

$$Y = \frac{\frac{fc}{a} - g}{\frac{bc}{a} - d} = \frac{ag - fc}{ad - bc}$$

Substituting this value for y in $ax + by = f$, we obtain

$$ax + b \frac{ag - fc}{ad - bc} = f \Rightarrow ax = \frac{f[ad - bc] - b[ag - fc]}{ad - bc}$$

Hence

$$x = \frac{f[ad - bc] - b[ag - fc]}{a(ad - bc)}$$

Thus, in this case, there is a unique solution for the system of equation.

Case 3 $a \neq 0, c \neq 0$ and $ad - bc = 0$

We know

$$ax + by = f \sim x + \frac{b}{a}y = \frac{f}{a} \sim cx + \frac{bc}{a}y = \frac{fc}{a} \sim cx + dy = \frac{fc}{a}$$

Thus, our original system of equation becomes

$$cx + dy = \frac{fc}{a}$$

$$cx + dy = g$$

Clearly, the above system has infinitely many solutions if $\frac{fc}{a} = g$ (that is, the system represents same line)

and No solutions if $\frac{fc}{a} \neq g$ (that is, the system represents parallel lines)

Rest of the cases are easy to work out!

Examples

① Solve $2x + 3y = 12$ $a=2, b=3$
 $x + \frac{3}{2}y = 4$ $c=1, d=\frac{3}{2}$
 $f=12, g=4$

Solution $a \neq 0, ad - bc \neq 0$ and $\frac{fc}{a} = \frac{12 \times 1}{2} = 6 \neq g$.

Therefore there are NO Solutions.

Here, note that $x + \frac{3}{2}y = 4 \sim 2x + 3y = 8$. Thus the system of equations becomes

$$\begin{array}{l} 2x + 3y = 12 \\ 2x + 3y = 8, \end{array} \quad \text{which represents parallel lines}$$

Clearly, this system has NO Solutions

② Solve $2x + 3y = 12$ $a=2, b=3, f=12$
 $x + \frac{3}{2}y = 6$ $c=1, d=\frac{3}{2}, g=6$

Solution Here, $a \neq 0, ad - bc = 0$ and $\frac{fc}{a} = g$.

Therefore, both the equations are one and the same and thus there are infinitely many solutions.

③ Solve $2x + 3y = 1$ $a=2, b=3, f=1$
 $x + 3y = 2$ $c=1, d=3, g=2$
 $ad - bc = 3$ $ag - fc = 3$

Solution $a \neq 0, ad - bc \neq 0$

Therefore, the system has exactly one solution

$$X = \frac{f[ad - bc] - b[ag - fc]}{a(ad - bc)} \quad Y = \frac{ag - fc}{ad - bc}$$

Thus, $X = \frac{3 - 9}{6} = -1, Y = 1$
